

CLASS XI · LECTURE 9

Memory Units and a Paper I Drill

One period of units, then a full mixed drill in the shape of the paper

By the end of this lecture you can

- Convert among bit, byte, kilobyte, megabyte, gigabyte and terabyte using 1024.
- Separate RAM from ROM, and volatile from non-volatile.
- Attempt a mixed Paper I drill under the real section rules.

How memory is measured

Inside the machine the smallest unit is a **bit**, which is either 0 or 1. A **nibble** is 4 bits. A **byte** is 8 bits, enough for one character in the usual coding. After the byte, the exam uses steps of **1024**, not 1000.

Unit	Equals	A way to picture it
1 nibble	4 bits	Half a byte
1 byte	8 bits	One character
1 kilobyte (KB)	1024 bytes	About half a page of plain text
1 megabyte (MB)	1024 KB	A short piece of music, or a thick book of text
1 gigabyte (GB)	1024 MB	A film, or a few thousand photographs
1 terabyte (TB)	1024 GB	A large hard disk

ON THE BOARD The one-mark conversion

1 GB = 1024 MB. It is not 1024 KB and it is not 1024 TB. To go up one named step, divide by 1024. To go down one step, multiply by 1024. Example: 4 GB = $4 \times 1024 = 4096$ MB.

RAM and ROM

Primary memory is on the motherboard and is directly used by the CPU. **RAM** is Random Access Memory. The CPU can read it and write it. It is **volatile**: the contents are lost when power goes off. It holds the operating system, the open program and the data you are working on. More RAM usually means more programs can stay open without the machine slowing down.

ROM is Read Only Memory. It is **non-volatile**. The ordinary user does not write over it. It holds the startup program, the BIOS. Variants you may meet by name: PROM can be programmed once; EPROM can be erased with ultraviolet light and rewritten; EEPROM can be erased electrically. You only need those names if the question lists them. The essential contrast is RAM volatile and writable, ROM non-volatile and used for permanent startup instructions.

Static RAM is faster and more expensive and is used for cache. Dynamic RAM is cheaper and is the usual main memory; it has to be refreshed. Secondary storage (Lecture 2) is non-volatile and larger, but slower to reach than RAM. Do not call a hard disk RAM.

A mixed drill

Do this on paper in one sitting. Section A: all fifteen, about twelve minutes. Section B: all parts, one side of each OR only. Section C: both questions, one side of each OR. Then open the model answers and mark strictly. If a point is missing, go back to that lecture rather than re-reading the answer three times.

Section A

1 One gigabyte equals: A) 1024 KB B) 1024 bytes C) 1024 MB D) 1024 TB.

- 2 Which topology uses the least cable for a small lab? A) Mesh B) Bus C) Full star with a spare hub D) A separate cable between every pair.
- 3 SMTP is used for: A) Printing maps B) Sending email C) Describing a web page D) Scanning a cheque.
- 4 Demodulation converts: A) Digital to analog B) Analog to digital C) Source code to object code D) A file into a folder.
- 5 The register holding the instruction currently being decoded is the: A) PC B) MAR C) IR D) Address bus.
- 6 A virus is: A) Hardware B) Software C) A protocol D) A topology.
- 7 Two similar LAN segments are joined by a: A) Bridge B) Plotter C) Compiler D) Daisy wheel.
- 8 OSI has: A) 4 layers B) 5 layers C) 7 layers D) 8 layers.
- 9 Half duplex means: A) One direction only B) Both directions, not together C) Both directions together D) No transmission.
- 10 Which is a magnetic disk? A) DVD B) Fibre C) Hard disk D) RAM.
- 11 HTML is a: A) Mail protocol B) Markup language for web pages C) Type of ROM D) Bus.
- 12 Which memory loses its contents when power fails? A) ROM B) Hard disk C) RAM D) DVD.
- 13 A large engineering drawing is produced on a: A) Plotter B) Microphone C) Bridge D) Bus.
- 14 The translator for a whole high-level program, producing object code, is a: A) Plotter B) Compiler C) Repeater D) Hub.
- 15 BIOS stands for: A) Basic Input Output System B) Binary Internal Operating Software C) Bus Interface Of Storage D) Basic Internet Output Signal.

Section B — answer every part, one side of each OR

- i) Define information technology and give three abuses. **OR** Explain the use of a barcode reader at a checkout.
- ii) Differentiate hard copy and soft copy. **OR** Differentiate synchronous and asynchronous transmission.
- iii) Define a computer virus and name five antivirus programs. **OR** Which device prints a large map? Explain briefly.
- iv) What is the fetch cycle? **OR** Name the five components of data communication.
- v) Why is a compiler more efficient at run time than an interpreter? **OR** What is modulation?
- vi) Give four differences between impact and non-impact printers.
- vii) Which device lets two different networks communicate? Define it.
- viii) Differentiate analog and digital signals.
- ix) Differentiate broadcast and point-to-point transmission.
- x) Write the full form of any three: OCR, GUI, CRT, BIOS, SMTP.

Section C — answer both, one side of each OR

- 1 Explain the functions of an operating system. **OR** Explain the types of CPU register.
- 2 Explain communication media with examples. **OR** Draw the OSI model and state the job of each layer.

Answer key for the drill

Section A: 1 C, 2 B, 3 B, 4 B, 5 C, 6 B, 7 A, 8 C, 9 B, 10 C, 11 B, 12 C, 13 A, 14 B, 15 A.

Section B and Section C are answered in Lectures 1 to 8. Mark your paper against those model answers. A short answer needs the definition plus the examples the question asked for. A long answer needs a heading, a definition, and a point for every item in the lecture table, written as sentences.

EXAM TIP How to mark yourself

Give the mark only if the point is on the page. A correct idea written as one vague line scores the definition mark, not the explanation marks. If you missed the fetch-cycle order, redraw it tonight from Lecture 3. If you swapped bridge and gateway, rewrite that pair from Lecture 7 before you sleep.